ME498: Artificial Organs

Cryopreservation of Sperm, Oocytes, and Embryos

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1. Introduction

It is becoming recognized that infertility is a widespread health issue with significant individual, medical, and social consequences. Clinically, infertility is traditionally considered to be the failure to conceive following 12 months of regular unprotected intercourse (or premature analysis in the elderly and patients with risk factors). Although infertility is a term used in discussing an individual diagnosis, it also serves as a population trend manifesting itself in larger tendencies in family planning, delayed parenthood, chronic disease, cancer survivorship, as well as unequal access to reproductive care. Over the last few years, professional societies and government health agencies have stressed the fact that infertility is no longer a rare experience, and the ability to retain reproductive capability, whether by using television-mediated methods like cryopreservation, has become a leading issue in the contemporary medicine of reproductive care.

1.1 Recent Statistics of Infertility: International and National Patterns

On an international scale, infertility is a significant problem in proportion to the population. According to a 2023 report by the WHO (World Health Organization), it has been estimated that 17.5% of adults around the world experience infertility (1in 6) at some time in their lives. This estimate draws attention to the whole world to infertility as a common health issue. Furthermore, the WHO manages infertility in the context of access and equity; most of the public in need of treatment are faced with financial and structural issues that do not allow them to access these fertility services.

According to the Centers for Disease Control and Prevention (CDC) in the National Survey of Family Growth (NSFG), they identified that of current married women in the 15-49 years of age bracket, about 8.5% of them qualify as infertile. This statistic is a typical measure of the percentage of married couples facing problems with childbearing in the United States surveys,

since infertility is operationalized by the exposure of married women to pregnancy risk over a specific period. The broader reproductive difficulty indicators reported by the CDC include: 13.4% of women between 15-49 experience impaired fecundity (difficulty in getting pregnant and carrying a pregnancy to term), which is broader than the 12-month definition of infertility and is more comprehensive of biosocial and biological challenges overall.

The case of lifestyle changes influences domestic trends. Individuals are postponing having children due to education, work, economic stability, and timing of relationship, and these are directly related to age alteration in reproductive potential. It is ethically acceptable and can potentially assist people in preventing future infertility; however, according to the clinical organizations like the American Society of Reproductive Medicine (ASRM), when considering planned fertility preservation and oocyte cryopreservation, patients should be made aware of uncertainties, anticipated success, and age-related outcomes. This flexibility shows an understanding of why fertility preservation is not considered to be associated with medical emergencies.

1.2 Introduction to Cryopreservation

Cryopreservation is the cooling and preservation of cellular elements or tissue of biological organisms under very low temperatures – typically liquid nitrogen between approx -196 deg-c, to permit the cellular metabolism to effectively terminate, permitting biological material to be maintained. In reproductive medicine, cryopreservation is used in the preservation of sperm, oocytes (eggs), embryos, and, in certain cases, reproductive tissues. It is thought necessary due to the fact that cryopreservation alienates reproductive time and biological time: a person can store reproductive cells at one moment in life and bring them any time, when he/she is prepared to pay attention to a pregnancy, or he/she needs them due to some medical issues.

Assisted reproductive technology (ART) has evolved from the concept of cryopreservation. Based on the example, embryo cryopreservation provides flexibility in terms of scheduling the embryo transfers and reduces the need to develop and transfer embryos just on a single hectic circuit. The freezing of eggs by cryopreservation allows one to freeze their eggs without the need for a sperm donor, which aligns with the patient's autonomy and diverse family-building processes. The clinical outcomes of both fresh and cryopreserved oocytes in the proper setting were similar based on the outcome of ASRM post-cryopreservation of oocytes, which adds to the idea of the cryopreservation of oocytes as a clinical technology, but not experimental.

1.3 Significance of the Topic: Why protect biological materials and why it matters

Cryopreservation has a considerably important role in life-course planning and medical necessity in the field of reproductive medicine. The medical aspect of fertility preservation is essential to patients, as their treatment might impair the functionality of their reproductive system. Fertility preservation is pointed out as a crucial factor to be considered by patients with medical predispositions. In the case of post pubertal females, mature Oocyte cryopreservation is stated to be a well-developed procedure, and patient values and expected effects of intervention on fertility are to be taken into consideration. In the same fashion, fertility preservation by means of sperm cryopreservation in males (gonadotoxic therapies, surgeries or other conditions that put the production or quality of sperm at risk) has a long history.

In addition to acute medical requirements, the importance of cryopreservation has come about since most individuals put off having children. The possibility of postponing family building may be a sensible and effective personal choice, yet it is also connected to biological limitations, particularly with age, as oocytes decline both in number and in quality. ASRM ethical advice justifies planned oocyte cryopreservation as something permissible and possibly helpful, although informed advice is necessary regarding effectiveness, uncertainties, and long-term issues. Practically, this implies that cryopreservation not only serves as a kind of backup plan, but it is a measure that can ease time pressures, increase the number of reproductive choices, and assist in the volume of independence in terms of where and when individuals get parenthood.

Lastly, cryopreservation is important since it also affects actual clinical outcomes of ART. Quality post-thaw sperm, oocytes, and embryos can influence the fertilization process, embryo growth, and level of pregnancy. Although oocyte and embryo cryopreservation, in particular Cryopreservation by vitrification, has led to more successful outcomes and survival rates than the older techniques, cellular stress and possible injury are still induced by the process of cryopreservation. Consequently, the maximization of cryopreservation techniques, such as the formulation of cryoprotein-dilution solutions and the freezing/thawing regimens, is a relevant and sought-after field of study. This encourages further comparison of protocols (e.g., slow freezing vs vitrification), assessment of post-thaw cellular integrity and activity, and the creation of more standardized and safer protocols in support of medical fertility preservation and the planned reproductive choice.

2. Significance: Specific reasons for sperm preservation

Cryopreservation of spermatozoa (sperm) is part of a broader technique of Assisted Reproductive Technology (ART). In general, ART is a fertility treatment involving the laboratory handling of sperm, oocytes, and embryos to achieve pregnancy [1]. Some of the popular ART methods include In Vitro Fertilization (IVF), Intracytoplasmic Sperm Injection (ICSI), and cryopreservation. IVF method involves fertilizing mature egg and sperm in laboratory and transferring the embryos into the uterus. ICSI method involves injecting a sperm directly into an egg to facilitate fertilization. Lastly, cryopreservation is a new method used to freeze reproductive cells for future use. Cryopreservation is typically used for cancer patients, and those delaying childbirth.

Cryopreservation of sperm in particular can enable long-term storage of viable male gametes for future use while ignoring the detrimental effects such as age, medical treatment, and infertility of male. Sperm cryopreservation is useful for several clinical situations such as before infertility surgery or treatments that might impair fertility; during and after surgical infertility procedures; before cancer therapy; prior to treatments for non-malignant diseases; and premortem and postmortem sperm banking (for preserving fertility after death) [2]. Finding has also shown that post-thaw use of cryopreserved sperm in ART such as IVF and ICSI has enabled conception rates comparable to fresh sperm in many settings, particularly with ICSI where only one viable sperm is needed per oocyte.

Sperm bank is also a key factor in the success of cryopreservation in technology. In 1949, the first cryopreservation of sperm was conducted using glycerol [3]. In 1953, the first successful pregnancy was achieved with the thawed frozen sperm. As such, 1953 marks the year that sperm bank is a feasible option for the next step in scientific advancement. In 1964, the first therapeutic sperm bank for infertility was established in Iowa, USA, and Tokyo, Japan. Sperm bank focuses on three groups, partner planning fertility protection, male suffering from infertility, and recipients

donating sperm. Sperm bank operates by providing services such as semen analysis, sperm storage, genetic counseling, donor insemination, and IVF usage. The global sperm bank market has also been shown to be expanding driven by rising infertility rates, greater awareness of assisted reproduction benefits, IVF demand, supportive policies, and evolving family structures. As seen in figure 1, sperm bank has a dedicated strategies for artificial insemination, either with a husband's sperm, or donor's sperm. Lastly, sperm banking is recommended especially for men who are about to undergo gonadotoxic therapy, since chemotherapy and radiation can irreversibly damage the testicular germinal epithelium, reducing or eliminating natural fertility.

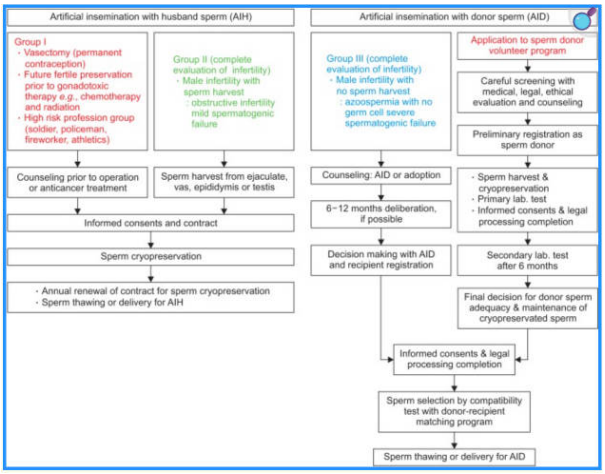


Figure 1: Strategy for artificial insemination program with husband or donor sperm.

Sperm cryopreservation starts with Lazzaro Spallanzani (see figure 2) when he first observed that snow-cooled human sperm became motionless in 1776 [4]. While not being a medical doctor, Spallanzani also helps in discovering medical achievements such as showing that fertilization requires both spermatozoa and ovum, performing IVF on frogs, and ICSI on a dog. In a standard cryopreservation procedure, the sperm is preserved in liquid nitrogen after treatment with cryoprotectants to protect cells during freezing and thawing [5]. Sperm is typically preserved using two methods: traditional freezing or vitrification. The traditional method used in ART requires a lengthy freezing procedure. Meanwhile, vitrification is a more rapid technique of sperm

cryopreservation since it does not require the removal of water. This method has the benefit of being faster for a "high cooling rate," but at the cost of increased contamination from liquid nitrogen and a lower sperm sample size. In 1963, long-term sperm cryopreservation using liquid nitrogen was demonstrated, and paving the way for sperm freezing to become known all over the world.



Figure 2: Portrait of Lazzaro Spallanzani

While generally is a wide-spread knowledge, it is worth defining spermatozoa to have better clarity of the sperm cryopreservation process. Sperm is a microscopic male reproductive cell produced in the testes and carry the genetic material of the male partner. Consisting of a head, midpiece, and tail (see figure 3), they are designed to swim and penetrate the egg for reproduction [6]. Millions of sperm are released during ejaculation within a fluid called semen. The head of the sperm contains the genetic material of the sperm cell within the cell nucleus and is covered in acrosome enzymes that can help in penetrating the layer of egg during fertilization. The body of the sperm contains mitochondria and is essential in providing energy for the sperm to travel to the egg. The tail of the sperm contains axoneme and consumes the energy from the mitochondria to propel the sperm towards the egg. Spermatogenesis is the process of formation of sperm cells in the testes. The testes are the male reproductive organs producing sperm and testosterone.

They contain tightly coiled structures called seminiferous tubules within which the sperm cells are produced.

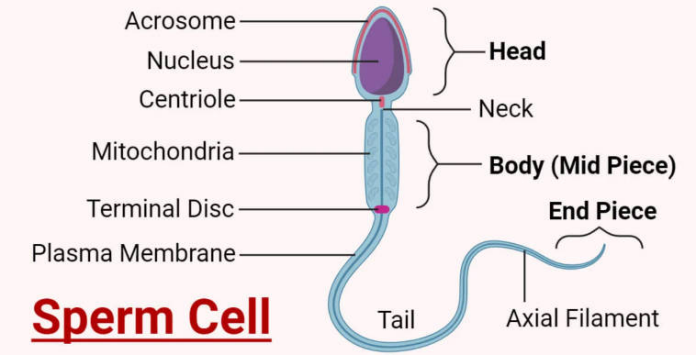


Figure 3: Anatomy of Sperm Cell

Sperm cryopreservation is typically recommended for people diagnosed with cancers. This is because chemotherapy and radiation, especially when aimed at the pelvic region, can nullify sperm production [7]. This is due to cyclophosphamide and other alkylating agents going after rapidly dividing cells, including sperm producing cells.

2.1 Cryopreservation Solution Protocol #1 (Glycerol-based vitrification):

Vitrification is a new technology still under lab trials used in sperm cryopreservation. Before starting the discussion on the main method used in sperm cryopreservation, we would like to explore new and upcoming technology researched in lab scale. Vitrification refers to the suspension of cells in appropriately high concentrations of CPA mixtures, which, combined with sufficiently high rates of cooling and warming, prevent ice formation in the cells and the surrounding medium during cooling or warming [8]. CPAs used in vitrifying cells are commonly the same type with those in rapid freezing, but the concentrations are much higher. Sperm

cryopreservation involves rigorous procedures to ensure the sperm undergo a proper freezing procedure. While a standardized sperm cryopreservation protocol is still yet to be established, many sperm cryopreservation techniques involve using glycerol-based solutions. The typical procedure used in a sperm bank (or any research conducted on the vitrification of sperm is as follow:

(1) Sperm collection, (2) Cryopreservation preparation, (3) Equilibration, (4) Freezing, (5) Storage, and (6) Thawing.

Sperm collecting involves collecting sperm specimen from a donor or a patient, ensuring to assess the sperm's volume, concentration, motility, and morphology to ensure it surpasses the threshold to undergo cryopreservation. The sperm is collected based on the following condition:

Patient Condition	Sperm Collection Method	What Makes Them Eligible for Sperm Banking?
Adult cancer	Frequent ejaculation	Cancer and related therapies damage the gonads and impair spermatogenesis
Adolescent cancer	Frequent ejaculation (when they are sexually mature), penile vibratory stimulation, and electroejaculation	Same as above
Cancer in prepubescent males	Testicular tissue freezing	Cancer treatment negatively affects spermatogenesis; testicular tissue extraction is a promising experimental procedure
Preoperative surgical procedure to treat or induce infertility	Frequent ejaculation	Bilateral varicocele ligation, prior to vasectomy
Non-malignant disease	Frequent ejaculation	Systemic stress may impair spermatogenesis; gonadotoxic therapies affect semen quality
Occupational risk	Frequent ejaculation	Exposure to harmful chemicals may decrease fertility or cause chromosomal damage in the germ cells; facing hazardous situations that may result in an accident that causes infertility
Posthumous sperm cryopreservation	Electroejaculation surgical removal of testes and epididymides	Performed at the time of brain death based upon the patient's will or family request
Sex change/ gender reassignment	Frequent ejaculation removal of testicular tissue	Hormone therapy damages spermatogenesis, and gender reassignment surgery sterilizes the male

Table 1: Sperm collection methods for various patients and donors [10].

A cryopreservation solution is prepared with a tris-based extender (buffer) solution (typically consists of tris, citric acid, or sugar) and high concentration of glycerol cryoprotectant (5-10%). The cryoprotectant (CPA) are divided into two types: those that can, and those that cannot, permeate cell membranes, both of which act by reducing the ice point of intracellular and

extracellular water, thus protecting cells from cellular mechanical injury caused by ice crystals during freezing and recovery. At the same time, the composition of a CPA can result in a more reasonable osmotic pressure of fluid, protecting the spermatozoa from damage caused by severe changes in osmotic pressure during freezing and recovery [8]. The first highly effective CPA was glycerol. Dimethyl sulfoxide (DMSO) was then founded to penetrate cells faster than glycerol and required a much lower concentration to completely prevent freezing damage. In addition, sugar was founded to be the first high-molecular-weight CPAs to be used as additives before freezing. In addition, disaccharides such as trehalose and sucrose are also now used more extensively. Notably, common additives include vitamin E, vitamin C, glutathione (GSH), melatonin (MLT), taurine, zinc/selenium, and nonenzymatic antioxidants such as L-carnitine. Enzymatic adjuncts include glutathione peroxidase (GPX), superoxide dismutase (SOD), and catalase (CAT). Herbal extracts, including genistein, rosemary, and rhodiola are also tested as additives. Using permeable cryoprotectants in the cryopreservation media increases the osmolarity, which fluctuates between 600 to 1000 mOsm/L [2]. In contrast, sperm vitrification media is isosmolar between 300 to 396 mOsm/L, which can be achieved with the use of nonpermeable cryoprotectants or a combination. Cryopreservation of sperm, free of permeable cryoprotectants, is important to maintain sperm fertilization potential such as capacitation, acrosome reaction, and the integrity of cytoplasmic and mitochondrial membranes.

The vitrification solution is then mixed with the sperm. More vitrification solution will gradually be added as the mixing continues to minimize osmotic shock. The mixing process occurs for 10-30 minutes at a temperature of 4°C to allow permeating glycerol cryoprotectants to enter sperm cells and replace the water within the cell. The permeation of glycerol allows for reduced ice crystal formation during the actual freezing process.

The treated semen is then rapidly plunged into a specialized containers such as Cryoloop containing liquid nitrogen to reach -196°C for 8-10 minutes to undergo rapid cooling at cooling

rates of 2, 000–10, 000 °C/min. Besides open systems like cryoloop, a number of closed carriers have been developed and applied in vitrification, including straw-in-straw, high security vitrification straw, Cryotip, VitriSafe, Cryopette, cryoLogic, Rapid-i, S3 system, and the S3 μS-VTF device [9]. The microstructure will be glass-like in appearance with no ice crystal forming inside the membrane. A high freezing and thawing speed can reduce the formation of ice crystal regions in and outside spermatozoa, the time of sperm exposure to osmotic pressure, and the damage to spermatozoa caused by the freezing and thawing process [8]. Currently, rapid freezing after adding CPAs comprising highly permeant chemicals is widely applied for the cryopreservation of single or small numbers of sperm [8].

The cooled specimen is then stored in a liquid nitrogen tank for long-term storage. Liquid nitrogen is commercially produced by compressing air and using fractional distillation [9]. Unless specifically requested, commercial liquid nitrogen provided by suppliers is not sterile. Therefore, it is a potential source of contamination as number of infectious agents (*Pseudomonas* spp*., Enterobacter cloacae, Staphylococcus sciuri, Acinetobacter calcoaceticus,* and *Flavobacterium* spp) can survive at cryogenic temperatures and impose increased risk of transmission and propagation of diseases.

If the specimen is needed for fertilization, the sperm is thawed rapidly in a warm water bath at 37°C to ensure ice recrystallization didn't occur. Rapid freezing gives superior post-thaw motility and survival rate [8]. A rapid warming rate is another necessary condition for the high survival rate of frozen spermatozoa. The survival of spermatozoa decreased when they were warmed at a suboptimal rate (10–100 times) after being frozen at an optimal rate (20°C–130°C per min). This is most likely due to the physical injury caused by ice recrystallization outside the cells during slow warming. The specimen is heated at an ultra-high rate of about 1×10^7°C per min, which is almost 100 times faster than conventional thawing. The exceedingly high rate seems to protect cells and embryos by preventing small ice crystal recrystallization formed during cooling.

Detailed Composition (Glycerol-Based Vitrification)

Component	Typical Concentration / Notes
Tris (buffer)	Maintain pH stability (190-210mM)
Citric acid (buffer)	Maintain pH stability (55-75mM)
Sugars (e.g., fructose)	Act as energy sources (45-65mM)
Ethylene Glycol / DMSO Glycerol (permeating cryoprotectant)	4–8% (v/v) (to prevent intracellular ice crystal formation and stabilize the plasma membrane)
Egg yolk protein (non-permeating protectant)	5–25% (v/v) (to protect membrane and minimize extracellular ice formation)
Electrolytes (Na+, K+, Cl-)	Maintain ionic balance
Antibiotics	Penicillin/Streptomycin (to prevent contamination)
Water (ultrapure)	100% solvent

Table 2: The exact content of the CPA is inferred from a patent by John L. Schenk [12]

Since vitrification still remains a new technology used to cryopreserve sperm, Table 2 serves as reference since it is taken based on slow-freezing cryopreservation solution. The cryopreservation solution patent was first applied on the year 2000. This cryopreservation solution will also be discussed again during in the slow-freezing cryopreservation method.

2.2 Cryopreservation Solution Protocol #2 (slow freezing):

Currently, the widely-used sperm cryopreservation method used in the industry is the slow-freezing method. To focus our discussion on the slow-freezing method used on sperm, the paper “Sperm Cryopreservation Today: Approaches, Efficiency, and Pitfalls” by Ozimic et al. will be our main topic of discussion [18]. This paper discusses on the background of cryopreservation, the general principles of cryopreservation, the cryodamage impact of cryopreservation on human sperm, factor causing cryodamage, cryoprotectant contents, slow-freezing technique, and studies on vitrification technique.

The most commonly used cryopreservation techniques are slow freezing and freezing on liquid nitrogen vapors. Meanwhile, the use of vitrification is still not accepted as clinically relevant. Although there have been many improvements, the ideal technique for achieving better post-thaw sperm quality continues to be a mystery. The first individual to discuss sperm banks was the Italian neurologist, anthropologist, and physiologist Paolo Mantegazza. In 1886, Mantegazza quoted: “It might even be that a husband who has died on the battlefield can fecundate his own wife after he has been reduced to a corpse and produce legitimate children after his death”. The possibility of cryopreserving human spermatozoa has existed for over 80 years now, and empirical methods are still used today. All cryopreservation techniques (slow-freezing, rapid freezing, and vitrification) are noted to have some differences between the protocols. The most significant difference is the speed of freezing and thawing, followed by the use of a cryoprotectant and its concentration. Studies have also agreed that most cryoinjuries occur due to freezing the cells at an incorrect freezing speed. If the speed is too fast, water is not removed quickly enough, and intracellular ice crystals are produced that can damage cell organelles. However, if the speed is too slow, the cells eliminate water rapidly and dehydrate. Cryodamage is typically not preferred because it can lead to (1) a decrease in sperm motility and viability, (2) a decrease in mitochondrial activity rates, (3) a decrease in DNA integrity, and (4) an increase in the production

of reactive oxygen species (ROS). As such, cryoprotective agents are developed to protect the cells during the freezing process and reduce the stress produced by cryopreservation techniques.

Cryodamage of Cryopreservation on Sperm

To retain several features necessary for egg fertilization, it is recommended to know the complete sperm physiology (such as motility, viability, acrosomal integrity, and DNA content) during cryopreservation and to consider the environment to which we expose spermatozoa during cryopreservation and how this environment is reshaped at different rates of cooling. In most cases, the most well-known cryoinjuries are ruined DNA integrity (DNA fragmentation, chromatin decondensation, and variations in acrosomal state), reductions in motility and viability, and changes in the plasma membrane's fluidity and integrity.

Plasma Membrane Fluidity and Spermatozoa Viability

Biomembranes are the main building blocks of cells which separate the outside and inside of an organism. Biomembranes are made of lipid bilayers with selected permeability, allowing which substances can enter and leave the cells. One of the components in biomembranes are protein. Protein serves in relaying signals between the cell and its environment, moving molecules and ions across the membrane, and involvement in the immune response, and enzyme activities. The othercomponents are lipids. Lipids are highly sensitive to temperature and work as energy storage molecules, chemical identifiers, and signaling molecules. Since cryopreservation involves sudden temperature changes, membranes are the primary site for cryoinjury to occur. Membranes are in a fluid phase under normal conditions, but when extreme temperature changes occur, the membrane's consistency breaks, and its components mix and lose conformation. Other than temperature-related injury, membranes can also destabilize due to large volume changes that are

associated with cryoprotectant, water movements, and osmotic stress. As such, during cryopreservation, membrane fluidity will decrease.

Motility

The determination of sperm motility is biologically and clinically important because the presence or absence of rapid, progressive spermatozoa gives us better prognostic information for ART. Sperm cells are rich in mitochondria because a constant supply of energy is required for their motility. Therefore, mitochondrial damage during cryopreservation processes is linked with a loss of membrane permeability. This is more evident from the fact that cryopreservation can impose between 25% and 75% chances spermatozoa to become nonviable or lose their motility.

DNA Integrity and Acrosome Integrity

The integrity of sperm DNA is a highly important factor for the success of ART. In study, it is noted that there is correlation between DNA damage and reduced fertilization rates, and increased risk of pregnancy loss. The acrosome is an important part of the sperm and is needed to remain intact until it reaches the egg. Studies have noted that acrosome damage induced by cryopreservation typically occurs because of a change in membrane fluidity, which is a consequence of elevated levels of ROS. Findings have also shown that sperm can maintain their DNA and acrosome integrity better when the temperature drops slowly.

Factors causing cryodamage

Oxidation stress (Oxidation process can lead to the formation of toxic products and consequently reduces sperm motility)

Osmotic Stress (During the freezing process, ice crystals start to emerge, and there is less water left in which the solutes dissolve, so the concentrations of extracellular solutes and CPAs increase, creating a hyperosmotic environment for the cells that can affect different cell mechanisms related to cell viability)

Temperature Stress (Cold shock can lead to the formation of ice crystals that are pure crystalline water and will have the ability to dissolve solutes)

Cryoprotectants and Antioxidants

Cryoprotectants are low-molecular-weight essential substances used for minimizing the stress produced by freezing and thawing, especially the stress induced by ice formation. Cryoprotectants alter the inside and outside of the cells to lower the freezing point of the solution and keeping the extracellular environment in the liquid phase at any temperature. Aside from limiting ice formation, cryoprotectant also enter a glassy state under a very low temperature which allows for slower molecular reaction and higher stability of long-term storage. There are two groups of cryoprotective agents (CPA), permeating CPA and non-permeating CPA. Permeating CPA penetrates the membrane, and non-permeating CPA does not.

The permeating CPA used most often are glycerol, ethylene glycol, dimethyl sulfoxide (DMSO), and 1,2-propanediol (PROH). These CPA move easily across the membrane and create an osmotic gradient which causes water to leave the cell and the cell to shrink. They also lower the freezing point and provide intracellular protection.

Non-permeating CPA have a high molecular weight which disables them from crossing the cellular membrane, so they generate an osmotic gradient outside the cell that causes water to leave the cell. Non permeating CPA are divided into disaccharides (sucrose and trehalose), polysaccharides (maltodextrin), and proteins (albumin).

The most common contents in CPA solution are glycerol, sugars, lipids, proteins, and salt. Glycerol's primary task is to protect the spermatozoa against thermal shock. This is achieved by influencing the cell membrane by acting on the stability and permeability of the lipid bilayer and its structure. The influence on cell membrane will change the cell's metabolism and associations with surface proteins. Other than that, glycerol is also used to prevent the formation of water crystals. Sugar is used to optimize the osmotic gradient and supply the spermatozoa with energy. Other than that, sugar can also increase the glass transition temperature and allow storage at lower temperatures. Protein (such as egg yolk) is supposedly used to improve the fluidity of the membrane and protect the integrity of the sperm.

Antioxidants are molecules that can reduce oxidative processes by eliminating the release of free radicals. Examples of antioxidants include vitamin E, glutathione (GSH), sericin, superoxide dismutase (SOD) and/or catalase (CAT), vitamin C, melatonin (MLT), selenium (Se), and some natural herbs (genistein, rosemary, curcumin, green tea extract, and oregano extract). Summaries of each antioxidants are as follow:

Table 3.

Studies investigating the effect of antioxidants supplemented with freezing media on cryopreserved spermatozoa.

Antioxidant Used	Method and Cryoprotectant Used for Cryopreservation	Patients Included in the Study	Results	References
Vitamin E	Freezing on liquid nitrogen vapor with glycerol as a cryoprotectant.	59 patients with asthenozoospermia and 38 normozoospermic patients (>20 × 10^6/mL and motility ≥50%)	Supplementing the cryoprotectant with VE significantly enhanced the total motility and progressive motility in normozoospermic as well as asthenozoospermic samples.	Kalthur et al., (2011) [82]
Melatonin	Freezing on liquid nitrogen vapor with glycerol as a cryoprotectant.	43 normozoospermic patients (>20 × 10^6/mL and motility ≥50%)	The results show that the supplementation of melatonin significantly increased motility and viability and decreased levels of intracellular ROS.	Karimfar et al., (2015) [83]
Sericin	Freezing on liquid nitrogen vapor with glycerol as a cryoprotectant.	51 normozoospermic patients (>20 × 10^6/mL and motility ≥50%)	The addition of sericin significantly increased sperm viability and total motility and decreased DNA fragmentation.	Aghaz et al., (2018) [84]
Oregano Extract (*Oregano vulgare*)	Freezing on liquid nitrogen vapor with glycerol as a cryoprotectant.	20 normozoospermic patients (>20 × 10^6/mL and motility ≥50%)	The total motility was significantly increased in frozen–thawed spermatozoa in comparison with the control group. The percentage of vital spermatozoa was also significantly higher.	Shiri et al., (2020) [85]
Green Tea	Freezing on liquid nitrogen vapor with glycerol as a cryoprotectant.	45 normozoospermic patients (>20 × 10^6/mL and motility ≥50%)	They found that supplementing the sperm-freezing media with GTE had a significant protective effect on human sperm motility and DNA integrity, but there was no significant change in the ROS level.	Alqawasmeh et al., (2021) [86]
Curcumin	Freezing on liquid nitrogen vapor with glycerol as a cryoprotectant.	23 normozoospermic patients (>20 × 10^6/mL and motility ≥50%)	In the curcumin group, progressive motility, sperm chromatin condensation, and DNA integrity significantly increased after the thawing process when compared with the control.	Karakus et al., (2021) [87]
Curcumin	Freezing on liquid nitrogen vapor with glycerol as a cryoprotectant.	60 normozoospermic patients (>20 × 10^6/mL and motility ≥50%)	The results showed that curcumin supplementation in a freezing medium was protective for human sperm parameters (increased total motility) and sperm DNA (decrease in DNA fragmentation).	Santonastaso et al., (2021) [87]

Figure 4: Antioxidants and its effect on sperm cryopreservation [18]

Detailed Composition (Slow-Freezing Cryopreservation)

Component	Typical Concentration / Notes
Tris (buffer)	Maintain pH stability (190-210mM)
Citric acid (buffer)	Maintain pH stability (55-75mM)
Sugar (e.g., fructose)	Act as energy sources (45-65mM)
Ethylene Glycol / DMSO Glycerol (permeating cryoprotectant)	4–8% (v/v) (to prevent intracellular ice crystal formation and stabilize the plasma membrane)
Egg yolk protein (non-permeating protectant)	5–25% (v/v) (to protect membrane and minimize extracellular ice formation)
Electrolytes (Na+, K+, Cl-)	Maintain ionic balance
Antibiotics	Penicillin/Streptomycin (to prevent contamination)
Water (ultrapure)	100% solvent

Table 3: The exact content of the CPA from a patent by John L. Schenk [20]

As seen in slow freezing (and vitrification), glycerol is the most widely used permeating CPA, as it readily enters sperm cells and displaces intracellular water, minimizing ice formation during the slow cooling process. Even low percentages (5–10%) significantly improve post-thaw motility compared with glycerol-free extenders. The full material masses for the cryopreservation solution are as follow [20]:

Components of Egg Yolk-Tris Extender

Glycerol:	4–8% vol/vol
Citric Acid:	55–75 mM
Tris [hydroxymethyl]aminomethane:	190–210 mM
Egg yolk:	5–25% vol/vol
Fructose:	45–65 mM
Tylosin:	25–100 µg/ml
Gentamicin:	200–300 µg/ml
Linco-spectin:	100–400 µg/ml*

*100–400 µg/ml lincomycin and 100–400 µg/ml spectinomycin

Figure 5: Egg yolk-tris extender material masses [20]

In a preferred embodiment of the invention, an egg yolk-Tris extender is used. However, Schenk also proposes multiple different material masses for the cryopreservation solution including an egg-yolk extender, and egg yolk TES-Tris extender, and a milk extender. In an alternative embodiment, an egg yolk-citrate extender is employed [20]:

Components of Egg Yolk-Citrate Extender

Glycerol:	4–8% vol/vol
Sodium Citrate:	60–80 mM
Egg yolk:	5–25% vol/vol
Tylosin:	25–100 µg/ml
Gentamicin:	200–300 µg/ml
Linco-spectin:	100–400 µg/mL*

*100–400 µg/ml lincomycin and 100–400 µg/ml spectinomycin

Figure 6: Egg yolk-citrate extender material masses [20]

In another alternative embodiment, an egg yolk-TES-Tris (“EY TEST”) extender is employed [20]:

Components of Egg Yolk TES-Tris Extender

Glycerol:	3–7% vol/vol
Tris [hydroxymethy-methyl]-2-aminoethanesulfonic acid:	140–170 mM
Tris [hydroxymethyl]aminomethane:	60–80 mM
Egg yolk:	5–25% vol/vol
Fructose:	5–12 mM
Tylosin:	50–150 µg/ml
Gentamicin:	400–600 µg/ml
Linco-spectin:	200–700 µg/mL*

*200–700 µg/ml lincomycin and 200–700 µg/ml spectinomycin

Figure 7: Egg yolk TES-tris extender material masses [20]

In yet another alternative embodiment of the invention, a Milk extender is employed [20]:

Components of Milk Extender

Homogenized Milk	90% (vol/vol)
Glycerol:	3–7% (vol/vol)
Fructose:	1.25% (wt/vol)
Tylosin:	50 µg/ml
Gentamicin:	250 µg/ml
Linco-spectin:	250/300 µg/ml*

*250–300 µg/ml lincomycin and 250–300 µg/ml spectinomycin

Figure 8: Milk extender material masses [20]

Cryopreservation Technique

In sperm cryopreservation, two standard freezing methods developed in the middle of the 20th century are still used today. The two freezing methods are programmable slow freezing and freezing on LN_2 vapors. Methods based on a slow cooling rate have roughly the same protocols, which involve adding CPAs before freezing, freezing the cells, thawing the cells, and removing the CPAs after thawing. Slow freezing methods are based on slowly lowering the temperature for 2-4 hours, especially in the zone that can be lethal for cells. Slow freezing will focus on balancing

the increasing dissolved substances and increasing ice formation. The slow temperature drop (0.5–1 °C/min) is controlled by an automated machine. The machine freezes the plastic straws before they are plunged into LN_2. During freezing on LN_2 vapor, the plastic straws are filled with the sperm sample and a cryoprotective medium, in a 1:1 proportion. The solution is then left at room temperature for 10 min. The solution is then placed 5cm above the LN_2 for 30 min. Lastly, the solution is plunged into the LN_2 for storage.

Slow freezing uses the maximum cooling rate that will result in adequate osmotic dehydration to maintain the cell water's chemical potential near an equilibrium with the chemical potential of the water in the partially frozen external medium [13]. The slow freezing is not ideal for cryopreservation of single or small numbers of spermatozoa. To add, conventional freezing method does not require automatic equipment [14]. The sample comes into direct contact with nitrogen vapors at −80 °C and then has been immersed in liquid nitrogen at −196 °C. This technique has major drawbacks such as being unable to control the temperature drop curve. Slow freezing consists of 2~3 steps and usually costs 2-4 hours to complete [15]. Firstly, spermatozoa accumulating by ejaculation or other techniques keep at room temperature for 10 min. Secondly, samples are gradually frozen from 20°C to 5°C at a rate of 0.5–1°C/min and then cooled from 5°C to −80°C at a rate of 1–10°C/min. Finally, samples are plunged into liquid nitrogen. In summary, slow freezing follows the same procedure as vitrification:

(1) Sperm collection, (2) Cryopreservation preparation, (3) Equilibration, (4) Freezing, (5) Storage, and (6) Thawing.

Sperm collection, cryopreservation preparation, equilibration, storage, and thawing process is the same for both vitrification and slow freezing. The only main procedural difference between vitrification and slow freezing is in the freezing step. Conventional freezing method does not require automatic equipment. The sample comes into direct contact with nitrogen vapors at −80 °C and then has been immersed in liquid nitrogen at −196 °C for 2-4 hours to undergo slow cooling at a cooling rate of 0.2–0.5°C/min. If done correctly, the microstructure will be glass-like in appearance with no ice crystal forming inside the membrane. A proper slow freezing procedure can reduce the formation of ice crystal regions in and outside spermatozoa, the time of sperm exposure to osmotic pressure, and the damage to spermatozoa caused by the freezing and thawing process [13]. Currently, slow freezing after adding CPAs comprising highly permeant chemicals is not widely applied for the cryopreservation of single or small numbers of sperm [13]. In a study with 33 human semen samples, an experiment-controlled study found both slow cryopreservation and vitrification had similar results [17]. Some study also concluded that conventional freezing resulted in higher motility and viability, while spermatozoa undergoing

vitrification were healthier regarding morphology and had fewer defects of sperm head, midpiece and tail.

Lastly, freezing is performed in liquid nitrogen vapors. The cooling rates also must be carefully chosen in order to avoid damaging the sperm [19]. If sperm are cooled slowly then they will dehydrate and intracellular ice formation will be avoided, but they will be damaged by the high hyper-osmotic conditions created while the ice is growing. Fast cooling rates are also damaging because the water cannot leave the cells fast enough and it will supercool, which increases the probability of intracellular ice formation. The cooling rate also depends on the concentration of CPA. During the thawing process, the warming rate depends on the freezing rate, on the type of container used to load the sperm, and on the concentration of CPA. In addition, the freezing diluent is usually removed before using the sperm for ART. This is done by ensuring that the glycerol is removed from the sperm safely in a glycerol-free solution since a diluent containing no glycerol will induce osmotic stress to the cells. To alleviate this problem, the sperm should be diluted gradually using several steps with sufficient time for the sperm to equilibrate to each new osmotic condition.

In comparison, vitrification will have the semen mixed with freezing mediums in a Cryoloop machine and then plunged directly into the LN_2. Due to the rapid cooling, vitrification requires high concentrations of CPA. However, the main issues occurs when sperm is not capable of tolerating high concentrations of CPA at high speeds due to its osmotic fragility. As such, the main issue with vitrification is its' inability to preserve large volumes of sperm and provide very low sperm survival rates.

In addition, studies have been conducted to compare the effectiveness of slow freezing in comparison to vitrification. The results showed higher sperm motility and viability in the samples using slow freezing method in comparison to the vitrification method, but DNA integrity was not

found to be significantly different between these two methods. An average reduction in motility of 40% compared with the control group was found, but there was no significant difference between the vitrification and vapor freezing. However, some studies have noted to achieve the opposite result, finding that motility and vitality were higher after freezing using the slow-freezing method than vitrification, but vitrification provided better results with respect to morphology.

2.3 Results for Sperm:

Principles and Mechanisms

Feature	Vitrification	Slow Freezing
Cooling rate	Ultra-rapid	Slow and controlled
Ice formation	Avoided (Ideally)	Minimized extracellular ice
CPA concentration	High	Low
Equipment needed	Automatic	Manual programmable freezer

Table 4: Comparison between vitrification and slow freezing

Advantages and Disadvantages

Method	Advantages	Disadvantages
Vitrification	Minimal intracellular ice, rapid processing, often higher post-thaw motility	High CPA toxicity risk if on low concentration, high osmotic shock risk if on high concentration, requires precise handling
Slow Freezing	Established protocol, lower CPA toxicity, well documented	Ice crystal formation risk, slower process

Table 5: Advantages and disadvantages for vitrification and slow freezing

Cost of Cryopreservation

While sperm banking is a brilliant solution, the typical cryopreservation procedure can go for around [21]:

- Consultation and initial testing: $150-$300
- Freezing and processing: $500-$1,200
- Annual storage: $250-$500 per year

Success Ratios in Practice

The effectiveness of sperm using slow freezing and vitrification results vary by lab protocol, cryoprotectant composition, and initial semen quality. Post-thaw survival is typically measured by sperm motility, membrane integrity, and viability. It is found that there were no significant differences on sperm progressive motility, viability and normal morphology rates between groups regarding to basic characteristics [1]. This can be shown in the following table:

Table 1. Basic characteristics of study population (n = 105).

Characteristic	Total		Progressive motility (%)	Viability (%)	Normal morphology (%)
	n	%			
Total	105	100	25.07 ± 9.01	81.90 ± 5.90	3.13 ± 1.61
Age					
≥ 35 years	37	35.2	24.45 ± 7.76	81.59 ± 4.96	3.27 ± 1.45
< 35 years	68	64.8	25.40 ± 9.66	82.06 ± 6.39	3.06 ± 1.70
Mean ± SD	33.28 ± 6.15				
P value			0.61	0.70	0.52

Table 6: Comparison of cryopreserved sperm for people below and above 35 years old

In a test conducted, a washing process was used on sperm sample and is shown to have an increased sperm motility parameter before cryopreservation [1]. After the frozen-thawed process, sperm motility significantly decreased compared with fresh groups. Cryopreservation by conventional freezing showed sperm motility parameters higher than those of vitrification groups. In vitrification groups, sperm motility parameters in washing sample were significantly higher than those of unwashed samples. However, there is no significant difference between washed and unwashed in conventional freezing groups. It was also founded that the washing process significantly increased the average percentage of normal morphology sperm. To note, sperm washing also significantly reduced the average percentage of sperm defects in fresh samples. The test between washed and unwashed sperm before the cryopreservation process was initiated, the following post-thaw result was proven:

	Sperm parameters				
Group n = 105	Motility (%)	Viability (%)	Normal morphology (%)	Head defect (%)	Midpiece and tail defect (%)
Fresh					
Unwashed (FU)	25.75 ± 9.01	81.90 ± 5.90	3.13 ± 1.61	92.50 ± 4.44	64.50 ± 4.30
Washed (FW)	29.49 ± 9.68[a]	86.24 ± 5.88[a]	3.66 ± 1.51[a]	91.47 ± 3.76[a]	64.10 ± 4.38[a]
Conventional freezing					
Unwashed (CfU)	14.76 ± 6.16[a]	53.10 ± 9.65[a]	1.36 ± 1.02[a]	94.67 ± 3.41[a]	68.48 ± 4.31[a]
Washed (CfW)	14.91 ± 6.25[b]	55.46 ± 8.00b, c	1.43 ± 0.94[b]	94.24 ± 3.54b, c	68.40 ± 4.41[b]
Vitrification					
Unwashed (VitU)	8.67 ± 5.31a, c	38.99 ± 8.62a, c	1.95 ± 1.08a, c	93.15 ± 3.65[c]	67.92 ± 4.41a, c
Washed (VitW)	9.78 ± 5.33b, d, e	41.35 ± 9.10b, d, e	1.87 ± 1.09b, d	92.95 ± 3.50b, d	67.27 ± 4.29b, d, e

Table 7: Comparison between motility and viability between washed and unwashed sperm

In general, conventional slow freezing is shown to have a higher survival rate (60% survivability) than the novel vitrification method (40% survivability). The conventional slow freezing of human sperm is noted to be entirely feasible and better at saving sperm motility and viability. Meanwhile, the vitrification protocol allows superior results in terms of normal morphology and sperm defects.

Fertilization, Pregnancy Outcomes, and Health of Offspring

In a test conducted, two clinical reports have been published on the successful use of human semen stored for longer than 20 years. Live births have been reported from in vitro fertilization with semen stored for 21 years and from IUI with semen stored for 28 years [23]. As such, the hypothesis that frozen cells stored suffer little or no damage during extended storage in liquid nitrogen may be proven valid. A study using a 40 years-old cryopreserved sperm concluded the sperm is able to deliver two healthy girls at the gestational age of 38 weeks. The babies weighed

2,693 and 2,948 g and have exhibited normal development to date. Considering the decrease in sperm fertilization power induced by cryopreservation, it can be easily inferred that intrauterine insemination and conventional *in vitro* fertilization (IVF) with frozen-thawed spermatozoa result in lower pregnancy rates compared with insemination with fresh sperm [24]. However, the outcomes are different for intracytoplasmic sperm injection (ICSI), because this procedure requires only a small number of motile spermatozoa for successful fertilization. Overall, the analysis shows that pregnancy rates from cryopreserved sperm depends on methodology, and live birth rates are similarly variable.

Studies reported that sperm cryopreservation causes an increase in sperm DNA damage [22]. However, sperms displaying altered genome and DNA damage are able to achieve fertilization, thereby posing concerns about the transmission of abnormal genetic material to the offspring. While the study noted concerns in small increase in DNA fragmentation post-thaw, there is no consistent evidence of increased genetic or phenotypic abnormalities in offspring conceived using cryopreserved sperm compared to fresh.

2.4 Conclusion

Vitrification and slow freezing are both effective methods of sperm cryopreservation. Slow freezing remains widespread due to its reliable and well-documented use of glycerol as a cryoprotectant, while vitrification offers an insight for a quicker processing and potentially better preservation of sperm motility and viability. Decision-making in clinical practice depends on laboratory capabilities, cost, and specific patient needs. Future research continues to optimize CPA combinations and protocols toward maximizing post-thaw outcomes with minimal toxicity.

3. Significance and Challenges: Oocytes

Similar to sperm, the cryopreservation of oocytes is also a part of the ART technique for preserving life. As a matter of fact, there are a lot of similarities between the cryopreservation of sperm and oocytes. More specifically for females, however, oocyte cryopreservation is important for those who are going through procedures/illnesses that can harm their health, eggs, reproductive system, or who just want to freeze their healthiest eggs earlier in life [1].

As science and technology has progressed, the feasibility and accessibility of cryopreserving oocytes have increased. This meant that the original challenges of oocytes have gotten better. One of which was the challenge of the oocyte meiotic spindle. During meiosis, the spindles attach and organize chromosomes, however if it is at the wrong temperature, the spindles do not work. Since the oocytes are meant to stay at 37℃, if the temperature drops between 25℃ - 4℃ (called the danger zone), the oocyte's spindles begin to become disorganized and the chromosomes inside the oocyte begin the drift off and disperse. Therefore, to overcome this issue, the cooling technique must be fast enough so that the oocyte does not stay in this temperature range for too long. It also should be noted that spindles can rebuild if they have not been damaged too much, but this is not a 100% success [2].

Another challenge that has been overcome is the small surface area to volume rate that oocytes have. Since oocytes are one of the largest cells in the body (~120 μm), the techniques used to cryopreserve sperm cannot be copied over to oocytes. While the surface area increases by r^2, the volume increases by r^3. The ratio of surface area to volume is 0.05 (while for sperm it is 4.3) which makes freezing a more difficult challenge because it is not as simple (as sperm) to get cryoprotectant agents (CPAs) in and water out fast enough to prevent damage to the oocyte [3].

As water is being removed and replaced with CPAs, the extracellular environment becomes more concentrated, making the oocytes want to equilibrate and lose water. This causes the cell to shrink. It is also not good to rapidly remove water and then rapidly inject CPAs since this can cause damage to the oocytes as well [4].

Lastly, researches of oocyte cryopreservation had to overcome the challenge of zona hardening. The zona on an oocyte is a part of the extracellular matrix, acting as a membrane wall for the oocyte. When a sperm reaches the zona, the zona identifies it and lets it bind to the oocyte. Then, the zona will harden and in a way “closing off” the oocyte so that multiple sperm do not fertilize it. However, some of the CPAs (such as DMSO and Ethylene Glycol) that are used to cryopreserve the oocyte increase the intracellular calcium. This change is comparable to the increase that is caused by fertilization, tricking the zona into hardening. If this happens, the oocyte can no longer be fertilized and requires intracytoplasmic sperm injection (ICSI). Although this is common in vitrification, different amounts of calcium can change how much effect this has. This graph in Figure 2 shows the effect that calcium has on zona hardening (for mice). In the top (a), it shows that the zona hardening rate with calcium takes significantly longer than without calcium. As a result, the sperm chances of penetrating the oocyte and fertilizing it decrease to more than ⅙ of the non-calcium vitrification [6].

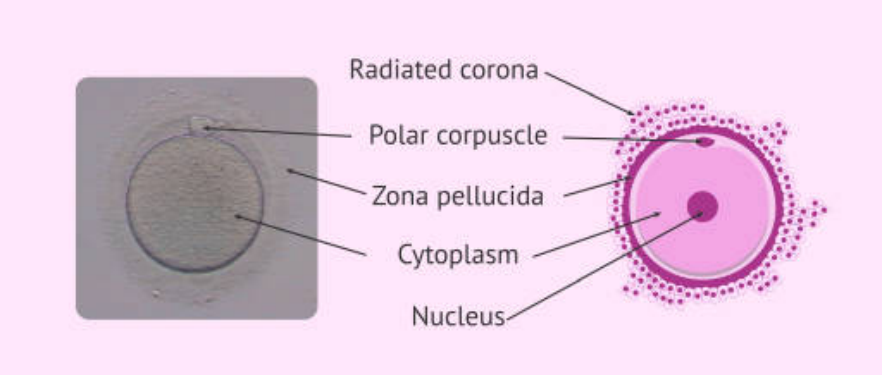


Figure 1: Anatomy of Oocyte [5]

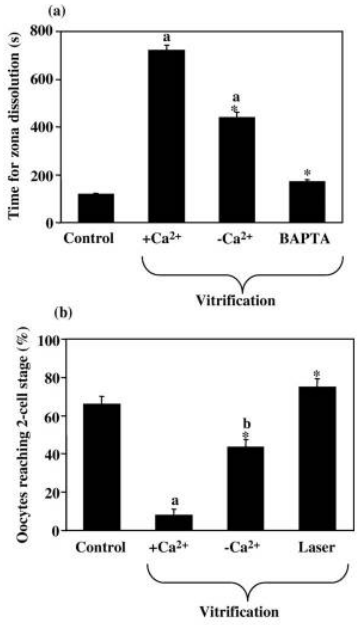


Figure 2: The Effect of Calcium on Zona Hardening [6]

3.1 Cryopreservation Protocol #1 - Vitrification

Freezing

The first of the two methods to cryopreservation of oocytes that will be covered in this paper is vitrification. Vitrification is a technique that rapidly freezes the oocytes to avoid crystallization. Cooling happens at a rate of -2500℃ to “outrun” ice crystal formation. Warming must be even faster than cooling to prevent the recrystallization of ice when it gets between the -150℃ to -50℃ temperature range [9].

Before the oocyte is frozen to be cryopreserved in vitrification, it must go through two different solutions to make sure that the composition of the oocyte is correct and prepared to undergo rapid freezing. The two solutions are the equilibrium solution (ES) and vitrification solution (VS). These

two solutions are meant to slowly introduce the CPAs into the oocyte to prevent osmotic shock if rapidly added in. The ES starts it off by slowly dehydrating the oocyte and filling it with CPAs while the VS tops up the CPA concentration and removes excess water. The solutions are meant to prevent ice crystal formation during the rapid freezing. After the solutions are prepared, the main structure to vitrification is as follows:

1. Put the oocyte in the equilibrium solution for 6-10 minutes
2. Place the oocyte into the vitrification solution for ~60 seconds to get the oocyte prepared for vitrification
3. Immediately after, plunge the oocyte in liquid nitrogen for ~80s
4. Store the oocyte in -196℃ temperature

These steps must be done immediately after one another to prevent the oocyte from staying in dangerous room temperature and overexposure to solutions for too long [9].

The solutions that the oocytes are placed in are consisted of the following:

- **ES Solution*:** 7.5% EG + 7.5% DMSO buffered in medium (M199 with 20% DSS)
- **VS Solution*:** 15% EG + 15% DMSO + 0.5 M trehalose + 20% DSS
- Liquid Nitrogen Plunge

*from source [10]

To better understand why each of the components are used in the solutions, here is a breakdown on what each is meant to do:

- **Ethylene Glycol (EG)**:** This is one of two CPAs that are introduced into the oocyte. The CPA permeates the membrane and replaces the water in the oocyte. EG is used in the equilibrium solution (ES) because it permeates quickly. Typically, EG is paired with

DMSO, another CPA, to build up the ES. This is meant to gradually be introduced into the oocyte so that the oocyte and other cells do not die from cytotoxicity that is suddenly introduced.

- **DMSO**:** DMSO is the second of the two CPAs that get introduced into the oocyte. This pairs well with EG because EG quickly permeates the oocyte while the DMSO has a good lower glass transition temperature, making it better at inhibiting the ice nucleation. Alone, DMSO is too toxic for the oocyte.
- **Protein Supplement (HSA/DSS):** Meant to maintain osmotic pressure and reduce extreme volume change during the addition and removal of cryoprotective agents (CPAs). Protein supplements also play a role in protecting the membrane and preventing the oocytes from adhering to plastics or micropipettes.
- **Buffered Solution (M199/HEPES-buffered HTF)***:** Important to stabilize the pH of the oocyte and balance the ions during the addition of CPAs. This makes sure that the oocytes do not swell up or acidify.
- **Sugar (Trehalose or Sucrose)**:** A sugar is typically added to the ES and VS to help draw out water from the oocyte via osmosis. They also help increase the viscosity which helps prevent ice crystallization. The sugar is extracellular (non-permeating).

**from source [11]

***from source[18]

<u>Thawing</u>

After the oocyte has been stored and is ready to be used, the egg must be thawed following the vitrification thawing protocol to ensure that the oocyte does not recrystallize when going through the -150℃ to -50℃ temperature range. Here is the process:

1. Immediately after removing the oocyte for storage, place the oocyte into the thawing solution for ~1 minutes.
2. Next, the oocyte will be put in a dilution solution for ~3 minutes.
3. Then, the oocyte will go into a washing solution for ~5 minutes.
4. Another wash will happen after the first wash.

[12]

Here is a more broken down version of the solutions:

- **Thawing Solution (TS)*:** 1.0M Sucrose + M199 + 21mM HEPES + 20% DSS at 37℃. This temperature is to ensure that the oocyte rapidly warms and passes the danger zones. The thawing solution has high sugar and protein content to help bring the oocyte back to temperature.
- **Dilution Solution (DS)*:** 0.5M Sucrose + M199 + 21mM HEPES + 20% DSS. The dilution solution has intermediate sugar concentration and allows the cell to re-equilibrate by removing CPAs and fill back with water.
- **Washing Solution (WS)**:** HEPES-Buffered M199 + 20% HSA/DSS. The washing solution is meant to remove any excess CPAs that are in the oocyte.

*from source [19]

**from source [13]

3.2 Cryopreservation Protocol #2 - Slow Freezing

Freezing

The second type of cryopreservation is slow freezing. Slow freezing, as the name suggests, is

slowly bringing the oocyte to cryopreservation temperature. This is meant to slowly introduce the coldness into the oocyte and dehydrate the oocyte to bypass ice formation rather than bypass it (vitrification). Similar to vitrification, slow freezing requires the oocyte to first be introduced in an equilibrium solution, a freezing solution, then finally plunged in liquid nitrogen. The ES is meant to introduce CPAs into the oocyte, the FS is meant to prepare the oocyte for freezing, and finally the liquid nitrogen plunge is meant to fully cryopreserve the oocyte. To more fully break it down the steps are as follows:

1. After the solutions are prepared, the oocyte will go into the ES solution for ~10 minutes
2. The oocyte is then transferred to the freezing solution for ~1 minute
3. Once prepared, the oocyte then is slowly freezed in a temperature controlled freezer (0.3-1C/min) until it reaches -150℃.
4. Immediately after, the oocyte is plunged into liquid nitrogen.
5. Finally, the oocyte should be stored in liquid nitrogen at -196℃.

[14]

The solutions are composed of the following:

- **Equilibrium Solution (ES):** HTF1023 Medium + 15% Serum Protein Substitute (SPS) + 1.5 M Propanediol (PROH), Room Temperature.
- **Freezing Solution (FS):** HTF1023 Base Medium + PROH + 0.2 M Sucrose, Room Temperature.
- **Slow Freezing:** The oocytes are loaded into a straw and put in a temperature controlled freezer. This slowly drops the temperature of the oocyte to -150℃.

[15]

The concentration of sucrose in the ES and FS impact it's ability to survive and it is important to have the right balance. There is a trend between increased amounts of sucrose and increased survival in thaw. When tested, 0.1 M sucrose had a ~24% survival rate after thaw while 0.3 M sucrose had a 71% survival rate after thaw [16].

Warming

The steps to thaw slow frozen oocytes varies slightly from vitrified oocytes because of the composition of the oocyte from the solutions. For slow frozen oocyte, the oocyte must be thawed first, then placed in diluted solutions to bring it back to equilibrium. The steps are as follows:

1. Oocytes are removed from the cryopreservation and air warmed for ~30 seconds.
2. They are then placed in a water bath (31℃) for ~60 seconds. This step is to remove any ice.
3. First dilution solution: HTF1023 + 15% SPS + 0.5 mol/L sucrose, 37°C, 10 min.
4. Second dilution solution: HTF1023 + 15% SPS + 0.3 mol/L sucrose, 37°C, 10 min.
5. Third dilution solution: HTF1023 + 15% SPS, 37°C, 7 min.

[15]

3.3 Comparison and Results

There are two main methods that are used to cryopreserve oocytes, vitrification and slow freezing. Slow freezing was originally the main method until vitrification was discovered. This table compares the two methods [8].

Summary Table

Feature	Slow Freezing (Controlled-Rate Freezing)	Vitrification
Cooling Rate	Slow, controlled (e.g., 0.3-1°C/min)	Extremely rapid (e.g., thousands °C/min)
Ice Formation	Extracellular ice formed; minimizes intracellular	No ice crystals formed; glassy state
CPA Conc.	Lower (e.g., 5-15%)	Higher (e.g., 30-60%)
CPA Toxicity	Lower concern	Higher concern
Equipment	Programmable freezer	Rapid cooling setup (e.g., direct plunge)
Mechanism	Dehydration, extracellular ice	Glass transition

Slow freezing is a method that freezes the oocyte in a slow and controlled manner. This allows for ice crystals to form which can be detrimental to the oocyte. However, slow freezing requires a lower concentration of CPAs, which in turn, means that it is less toxic. On the other hand, vitrification is a cryopreservation technique that rapidly freezes the oocyte as quickly as possible to avoid ice crystallization. This requires a higher CPA concentration and therefore is more toxic.

Of the two methods, slow freezing has become outdated by vitrification. As the technology gets better and better, vitrification techniques have gotten more advanced and the rate for the oocytes surviving cryopreservation has increased tremendously. In a study with over 50000 oocytes tested, the rate that oocytes survived after vitrification was ~92.5% compared to only ~60% for slow freezing [7]. The reasons for this was discussed earlier in the challenges to cryopreservation. Although slow freezing uses less toxic CPAs, it puts the oocyte at risk during the freezing process by bringing it to the danger zone of -150℃ to -50℃ for prolonged periods of time. At these temperatures, the oocyte is more at risk of crystallizing and dying. It also brings the oocyte in the danger temperatures of 25℃ to 4℃ when letting it rest in the ES that is at room temperature. These risks make slow freezing less effective, and it is shown in the data. Because vitrification rapidly freezes the oocyte, it bypasses the danger zone for spindle damage and the danger zone for ice crystal formation. It is for this reason that vitrification is favored over slow freezing.

Through a study with oocytes, when looking at data between vitrified and slow frozen oocytes, it is shown that vitrified oocytes have a higher survival rate. For vitrified oocytes, the survival rate is >90% while slow frozen oocytes only have a >60% survival rate.

When looking at the rate of pregnancy, after sperms are introduced to vitrified oocytes, there was a >75% fertilization rate. Then, in cases where fertilization was successful, there was then a 32-65% pregnancy rate [17]. These rates are only slightly lower than the rate using fresh oocytes with the pregnancy rate for the vitrified oocyte being just about the same as that of fresh oocytes. This shows that the vitrified oocytes are very effective. It is also important to note that from the vitrified oocyte pregnancies that have been successful, there were no signs of birth defects from babies.

Cryopreservation has come a long way, showing increased development in both techniques and tools used to assist. Of the two techniques, vitrification can take the clear lead in successful cryopreservation as not only is survival rates higher, but birth rates are on par with fresh oocytes. With many couples seeking to cryopreserve their offspring, vitrification is a reliable technique that they can turn to.

4. Significance (Embryos):

Embryo banking has become a massively important process for preserving reproductive potential for those experiencing infertility, whether it be due to cancer treatment, disease, genetic issue, etc. In in vitro fertilization (IVF), embryo banking refers to fertilizing retrieved oocytes to create embryos, culturing them to a defined stage (cleavage or blastocyst), and cryopreserving them for later transfer [3]. Those at reproductive-age affected by infertility are about 17.5% worldwide, highlighting just how many people are positively impacted by the use of fertility preservation. In the United States alone, as of 2022, 435,426 assisted reproductive technology (ART) cycles were performed on 251,542 different patients, with 184,423 being egg or embryo banking cycles in which all resulting eggs or embryos were frozen for future use [2]. Embryo banking has become a common practice, increasing success for IVF to occur while reducing the potential risks.

Main indication for human embryo cryopreservation in ART treatments.

Indication	Rationale for cryopreservation
Preimplantation genetic testing	Genetic assessment is facilitated by the opportunity to utilize cryopreservation to store embryos to be transferred in a future cycle, and to overcome the necessary time interval between blastocyst biopsy and genetic analysis (Coates et al., 2017)
Risk of ovarian hyperstimulation syndrome **(OHSS)**	When fresh embryo transfer cannot be performed due to the risk of exacerbating OHSS should pregnancy occur, embryos might be cryopreserved and used in a future cycle (Kawwass et al., 2015; Sciorio and Esteves, 2020)
Elective single embryo transfer (eSET)	The cryopreservation of surplus embryos is considered a valid method to reduce the number of embryos transferred during a fresh cycle and thus minimize the risk of multiple pregnancy, and reduces the need for repeated stimulation cycles [Johnston et al., 2014; Sciorio et al., 2019; Liebermann, 2021]
Preservation of fertility	In women with a stable partner about to go through gonadotoxic/chemotherapy treatments for cancer, there may be time in which to undergo a cycle of IVF and have blastocysts cryopreserved [Somigliana et al., 2015; Sciorio and Anderson, 2020]
Elevated progesterone or other conditions affecting endometrial receptivity, such as endometriosis, endometritis, hydrosalpinges, and fluid within the endometrial lumen	Elevated progesterone in the late follicular phase has a negative impact on pregnancy outcomes. Or other conditions and medical pathology that might affect fertility (Venetis et al., 2013; Groenewoud et al., 2018)

Figure 1. Different reasons as to why Assisted Reproductive Technology (ART) is used [3]

Following the first successfully performed birth using IVF in 1978, the process of embryo cryopreservation had become a large breakthrough because of its ability to allow embryos in one IVF cycle to be stored and used for later rather than using a fresh transfer every time. In the early stages of clinical cryopreservation, slow freezing was the main method and in 1983, Trounson and Mohr were able to perform a pregnancy through cryopreservation, and after thawing and transferring an eight-cell embryo, showing the possibility of a successful procedure [7]. Over the following decades, frozen embryo transfer (FET) has expanded from a successful experiment to become routine practice. In the late 1990s, the field had its second major breakthrough with the development of vitrification, inevitably replacing slow freezing as the main cryopreservation method due to its consistently strong post-warming embryo recovery and pregnancy outcomes, while also giving laboratories an easier step-by-step process to work through [2].

The task of embryo cryopreservation proves to be difficult due to it not being a singular cell, rather a multicellular structure with many different membranes and junctions that require great care and detail to ensure it survives. At the blastocyst stage, there exists a fluid-filled cavity, called the blastocoel, that can respond to drastic changes in water content, requiring extreme diligence when interacting with it. As temperature within the embryo drops, the main difficulty comes when controlling where water goes and how fast it moves, as too much water leads to problems during ice formation or recrystallization during warming, and too much water drained causes the embryo to endure extreme shrink-swell cycles, ultimately leading to injury. Therefore, successful protocols must be centered around carefully timed solution exchanges, where dehydration is carefully done before it is cooled and rehydration is performed with the same diligence when warming.

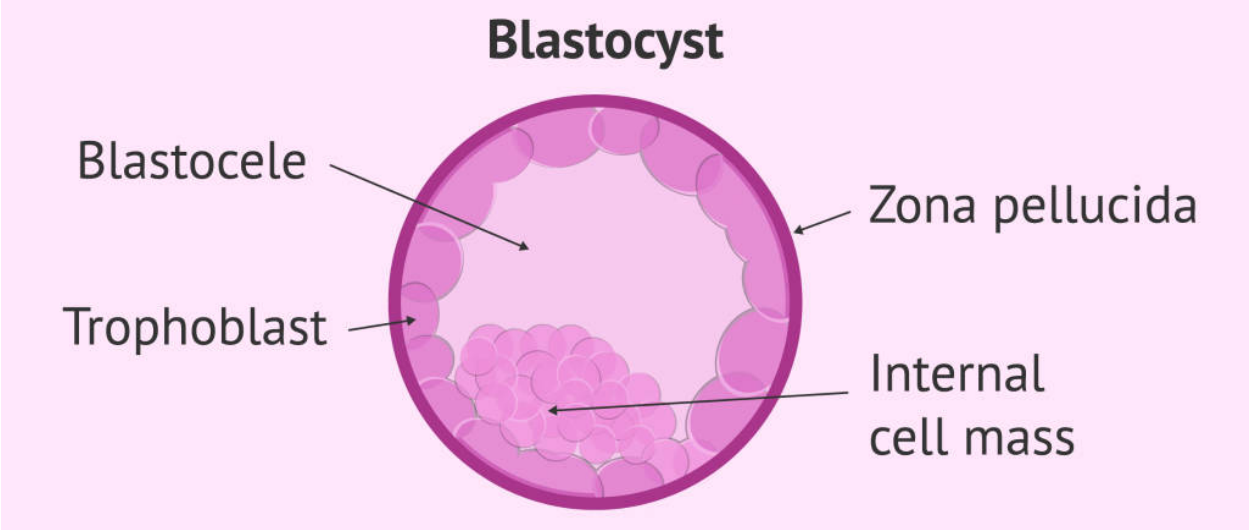


Figure 2. Anatomy of blastocyst [9]

As said above, in IVF, cryopreservation enables embryos to be stored long-term and transferred in a later cycle, which allows patients to attempt multiple pregnancies from a single retrieval without repeating ovarian stimulation. Achieving this successfully depends on preventing ice-related damage while also limiting the cryoprotectant toxicity and osmotic shock that can occur during the solution exchanges used for dehydration and rehydration. Vitrification is able to address the ice problem by using brief exposure to highly concentrated cryoprotectants and extremely rapid cooling so the solution solidifies into a glass-like state rather than forming damaging ice crystals [3]. When embryos are needed, they are warmed and moved through controlled dilution steps to remove cryoprotectants and restore water balance before further culture and/or transfer. For these reasons, solution composition and step timing are critical to successful cryopreservation. The next sections describe slow freezing and vitrification protocols in detail.

4.1 Protocol 1: Vitrification

Vitrification, often described as “non-equilibrium cryopreservation”, is able to successfully preserve the embryo through avoiding the temperature range where detrimental ice crystals would normally grow [3]. Instead of slowly freezing, embryos are exposed briefly to highly concentrated cryoprotective agent (CPA) solutions in very little volumes, then cooled and warmed at extremely high rates so the solution solidifies into a glass-like state rather than crystallizing into damaging ice. The key design features that vitrification has are (1) minimizing volume around the embryo so that the solutions are able to maximize cooling/warming rates, (2) using a mixture of penetrating CPAs (commonly ethylene glycol and DMSO) plus an extracellular osmotic agent (often sucrose or trehalose) that dehydrates the embryo and reduces the risk of intracellular ice, and (3) emphasizing that the warming phase is not “just thawing”. The warming solution must rapidly meet conditions where recrystallization can occur and must remove CPAs while also restoring water balance without causing osmotic shock. Because blastocysts contain a fluid-filled cavity (blastocoel), labs often use blastocoel collapse/shrinkage strategies before vitrification to reduce water volume and further lower the risk of cryoinjury.

TABLE 1.

Minimal concentration required to vitrify for some permeating cryoprotectants at a pressure of 1 atmosphere according to Fahy and colleagues 1984. PG, propylene glycol; DMSO, dimethyl sulfoxide; EG, ethylene glycol; GLY, glycerol.

Cryoprotectants	Concentration required to vitrify %/volume
DMSO	49–50
PG	43.5
EG	55
GLY	65

Figure 3. Minimal concentration required for different permeating cryoprotectants at a pressure of 1 atm [3]

Materials and Methods

The complete cryopreservation workflow for slow freezing uses both the vitrification, or freezing solutions as well as the thaw, or warming solutions that help to restore water balance and remove cryoprotectants after storage of the embryos. Every vitrification process is all centered around the same idea: short exposure to small volumes of highly concentrated solutions containing CPAs

One approach vitrifies embryos at either the cleavage-stage (day 3) or blastocyst stage (day 5/6) using a two-step exposure to penetrating cryoprotectants prior to rapid cooling in liquid nitrogen. In this protocol, embryos are loaded singly or in pairs onto a carrier for vitrification. All incubation is performed at 37°C, with embryos first held in VS1 for 2 minutes (3 minutes for blastocysts), then moved through three drops of VS2 for 15 seconds each before loading onto a carrier device for rapid cooling and storage in liquid nitrogen. This protocol can be implemented with different carriers, such as the Rapid-i or cryoloop [5]. The Rapid-i is a closed device that helps limit direct exposure to liquid nitrogen, while the cryoloop is an open device that places embryos on a thin film before being plunged into liquid nitrogen. For blastocysts, artificial shrinkage/collapse of the blastocoel using an ICSI needle can be performed to improve survival by reducing fluid volume [5]. After storage, the embryos are warmed and cryoprotectants are removed using a defined warming solution pathway that emphasizes stepwise osmotic recovery. One warming approach uses decreasing sucrose concentrations (0.25 M and 0.125 M) in Global medium with 20% SPS to gradually rehydrate embryos. Another warming protocol (using Cryolock® and Fujifilm NX warming media) applies a higher-to-lower sucrose dilution sequence and compares two step designs: a standard multi-step workflow versus a simplified single-step workflow. In the standard warming workflow, the blastocysts are exposed to the thaw solution (TS) at 37°C for 1 minute and then moved into dilution solution (DS) and wash solution (WS) with timed rinses, before transfer

through culture drops and return to incubation prior to transfer. In the single-step warming workflow, embryos undergo the same initial TS treatment, but the steps afterwards involving both the DS and WS are removed and embryos are instead rinsed through pre-equilibrated culture media and returned to the incubator prior to transfer.

Detailed Composition (electrolytes, nutrients, water balance)

Vitrification Solution #1 (VS1) is made up of 7.5% DMSO and 7.5% ethylene glycol (EG) in Global medium with 20% synthetic protein supplement (SPS), while Vitrification Solution #2 (VS2) contains 15% DMSO, 15% EG, 10 mg/mL Ficoll-70, and 0.65 M sucrose. Each chemical plays a specific role in enabling vitrification, as EG and DMSO act as penetrating cryoprotectants that permeate the embryo membrane and reduce intracellular ice formation during the rapid cooling process, sucrose acts as a non-penetrating osmotic agent that dehydrates the embryo before vitrification and supports controlled rehydration during warming, Ficoll-70 helps increase viscosity and supports vitrification in very small volumes, and SPS provides membrane stabilization when rapid solution changes occur. The solution compositions for the Cryolock®/Fujifilm NX warming protocol are explicitly defined: TS = 1 M sucrose + 20% DSS + gentamicin in M-199 HEPES-buffered medium, DS = 0.5 M sucrose + 20% DSS + gentamicin, and WS = 20% DSS + gentamicin in M-199 HEPES-buffered medium [10]. Overall, these vitrification protocols accomplish two goals that would normally conflict one another: using very high cryoprotectant concentrations and rapid cooling in order to avoid ice crystal formation, while also limiting the embryos' exposure time and using staged osmotic steps during the warming process to reduce the toxicity from cryoprotectant and osmotic injury.

Table 1: Embryo vitrification chemicals and uses

Component	Use

Ethylene glycol (EG) (permeating CPA)	A permeating cryoprotectant (often used with DMSO) to help embryos undergo rapid cooling by reducing intracellular ice formation risk.
Dimethyl sulfoxide (DMSO) (permeating CPA)	Common permeating CPA paired with EG; effective but requires careful handling because exposure to DMSO can cause toxicity/osmotic stress if not handled properly.
Sucrose (non-permeating osmotic agent)	Used in warming/thaw solutions, and some vitrification solutions to drive controlled dehydration/rehydration via osmotic gradients and help with safe CPA removal during warming.
DSS (Dextran serum substitute)	Serves as a macromolecular supplement (protein/dextran-based) used in warming solutions.
M-199 HEPES-buffered medium (base medium)	Base electrolyte/nutrient environment for TS/WS, balances pH.
Gentamicin (antibiotic)	Reduces contamination risk when handling.

Table 2: Vitrification (solutions + detailed workflow)

Section	**Step/Solution**	**Composition**	**Reason for Usage**
Solutions (composition)	VS1 (Equilibration Solution)	7.5% EG + 7.5% DMSO (v/v) (in base medium)	Used as the first CPA equilibration step before VS2
	VS2 (Vitrification	15% EG + 15%	Final high-CPA

	Solution)	DMSO (v/v)	exposure immediately prior to loading and cooling
	Warming Solution 1 (sucrose)	0.25 M sucrose	First CPA removal / rehydration step
	Warming Solution 2 (sucrose)	0.125 M sucrose	Second CPA removal / rehydration step
	NX Thaw Solution (TS) *(alternative warming system)*	1.0 M sucrose + 20% DSS + gentamicin in M-199 HEPES-buffered medium	TS exposure begins warming
	NX Dilution Solution (DS)	0.5 M sucrose + 20% DSS + gentamicin	DS is used in standard multi-step warming
Workflow (how to do it)	1. Choose embryo stage		Vitrify cleavage-stage (day 3, ~8–10 cell) or blastocyst (day 5/6) embryos
	2. Optional: collapse blastocyst		If blastocyst is expanded, perform blastocoel collapse prior to vitrification (method varies: needle/laser/etc.)
	3. Equilibrate in VS1	VS1 above	Place embryo into

			VS1 for 2–3 minutes (depends on stage of embryo)
	4. Expose to VS2	VS2 above	Move embryo into VS2 for 45 seconds
	5. Load onto carrier		Load embryo onto Rapid-i (closed) or cryoloop (open) with minimal fluid volume
	6. Cool & store (insert into bank)		Immediately place carrier into liquid nitrogen (LN_2, ~ -196°C) for storage
	7. Warm (remove from bank)		Remove carrier from LN_2 and begin warming immediately (warming rate is critical for avoiding recrystallization)
This step has multiple different ways of doing it, hence the 8A, 8B, 8C	8A. Standard sucrose warming (Rapid-i style)	0.25 M then 0.125 M sucrose	0.25 M sucrose for 5 min at 37°C → 0.125 M sucrose for 5 min at 37°C → base medium

			wash 5 min at 37°C, then culture
	8B. NX Standard warming (multi-step)	TS/DS/WS above	TS: 1 min at 37°C → DS: 4 min at room temperature → WS: two rinses, 4 min each → 5 drops culture media → incubate 1–4 hrs
	8C. NX Single-step warming	TS above + culture media	TS: 1 min at 37°C → rinse through 5 drops of pre-equilibrated culture media → incubate 1–4 hrs
	9. Post-warm assessment & transfer		Assess the survival/re-expansion (blastocysts) after recovery culture,

			proceed to transfer depending on clinic protocol

Results for embryos

Across both cleavage and blastocyst stages embryos, the post-warming survival was very high and clinical outcomes were similar across carrier implementations and simplifying the warming workflow (multi-step vs single-step) maintained strong embryo recovery and similar overall clinical performance. This reinforces the idea that the warming solution composition and step design are a major part of cryopreservation success, and serve just as important a purpose as the vitrification solution. Both how the embryo is able to be stored, and revived to be used for later use are critical for maintaining the embryos survival, and allow those who need to store embryos for later use success in their reproductive process.

4.2 Protocol 2: Slow Freezing

Before vitrification, slow freezing was the traditional embryo cryopreservation method that cools embryos at a controlled, gradual rate using lower cryoprotectant concentrations than vitrification, with the goal of reducing thermal shock while still limiting ice formation. However, because slow freezing passes through temperature ranges where ice crystals are able to form, outcomes are usually more sensitive to cooling rate control and intracellular ice injury. In this report, this section will be expanded to include a full slow-freezing solution composition and stepwise cooling schedule (cryoprotectant exposure, seeding/cooling rates, and thaw/dilution steps), so that the method can be directly compared to vitrification on survival and clinical outcomes.

Slow freezing is able to preserve the embryo by using controlled, stepwise cooling and lower CPA concentrations than vitrification, with the goal of letting water leave the embryo gradually as the

extracellular ice forms, reducing intracellular ice formation and limiting thermal shock. During a given slow-freezing process, embryos are first equilibrated in a permeating CPA (commonly propanediol/PROH for embryos, often paired with sucrose as an osmotic agent), loaded into straws, and placed into a programmable freezer where temperature is reduced in stages. The main characteristic step of slow freezing is manual “seeding” near subzero temperatures to initiate ice nucleation in a controlled way, after which cooling then continues at a slow rate through key temperature zones before there is a rapid drop to very low temperatures and storage in liquid nitrogen. After storage, embryos are then thawed and then dilution is used to rehydrate the embryo, remove CPA and restore water balance. Compared to vitrification, slow freezing is usually more sensitive to cooling-rate control and the time spent in temperature ranges where ice crystal growth can occur, which is why vitrification often performs especially well at the blastocyst stage where water content is higher.

Materials and Methods

This slow freezing protocol compares the normal two-step freezing process to a modified one step process that was originally used for improving human oocyte survivability, and is now being tested for cryopreserving cleavage-stage embryos using a commercial freezing kit in order to see if the modified process also improves cryosurvival for embryos. In the conventional approach, embryos are equilibrated at room temperature in F1 medium for 10 minutes, then transferred to F2 medium and loaded into cryogenic straws. In the modified one-step method, embryos are only dehydrated in F2 medium at ambient temperature for 10 minutes before it was loaded into cryogenic straws [5]. After loading, straws are placed in a programmable freezer starting at 23°C, cooled to −7°C

at −2°C/min, and then seeded. Cooling then continues at −0.3°C/min to −30°C, followed by rapid cooling to −150°C at −50°C/min, after which the straws are plunged into liquid nitrogen for storage [5]. After the freezing process, the embryos are then ready to be thawed when later used. In terms of thawing, the frozen straws are removed from storage and embryos are thawed rapidly by exposure to ambient temperature, then rehydrated through sequential incubation steps before being transferred into equilibrated culture medium at 37°C for assessment and eventual embryo transfer [5].

Detailed Composition (electrolytes, nutrients, water balance)

The commercial freezing kit that was used (manufactured by Irvine Scientific Inc.) consists of two solutions: F1 medium = 1.5 M propanediol (PROH) and F2 medium = 1.5 M PROH with 0.1 M sucrose [5]. The thawing kit uses a stepwise osmotic recovery sequence to remove cryoprotectant within the embryo and restore its water balance. Embryos are incubated for 5 minutes each at ambient temperature in T1 = 1.0 M propanediol + 0.2 M sucrose, then T2 = 0.5 M propanediol + 0.2 M sucrose, then T3 = 0.2 M sucrose, before they are then transferred back into the culture medium [5]. In terms of what these elements do for the slow freezing process, PROH is what penetrates the embryo, and serves as the cryoprotectant that reduces the intracellular ice risk during slow cooling, and sucrose is used as an osmotic agent that helps with both the dehydration before the freezing process and rehydration when the embryos are thawed for use.

Table 3: Embryo slow freezing components and uses

Component	Use
Propanediol (PROH) (permeating CPA)	A permeating cryoprotectant (often used with DMSO) to help embryos undergo rapid cooling by reducing intracellular ice formation risk.
Sucrose (non-permeating osmotic agent)	Included in F2 and thaw media to drive controlled dehydration before freezing and controlled rehydration and CPA removal during the thaw process.

Table 4: Slow Freezing (solutions + detailed workflow)

Section	Step/Solution	Composition	Reason for Usage
Solutions (composition)	F1 (Freezing Solution 1)	1.5 M PROH (propanediol)	Used for CPA equilibration in conventional method
	F2 (Freezing Solution 2)	1.5 M PROH + 0.1 M sucrose	Used prior to straw loading (both conventional and one-step method)
	T1 (Thaw step 1)	1.0 M PROH + 0.2 M sucrose	Stepwise CPA removal begins
	T2 (Thaw step 2)	0.5 M PROH + 0.2 M sucrose	Continued CPA dilution
	T3 (Thaw step 3)	0.2 M sucrose	Final osmotic recovery before return to culture

	NX Dilution Solution (DS)	0.5 M sucrose + 20% DSS + gentamicin	DS is used in standard multi-step warming
Workflow (how to do it)	1A. CPA loading (conventional two-step)	F1 then F2	F1: 10 min at room temperature → transfer to F2 → load into straws
	1B. CPA loading (modified one-step)	F2 only	F2 only: 10 min at room temperature → load into straws
	2. Start programmable freeze		Place straws into programmable freezer at 23°C
	3. Cool to seeding point	VS2 above	Cool to −7°C at −2°C/min
	4. Manual seeding		At −7°C, induce nucleation manually against straws for a

			few seconds (cotton swab pre-soaked in LN_2)
	5. Slow cool through critical zone		Continue cooling to −30°C at −0.3°C/min
	6. Rapid cool to deep subzero		Cool to −150°C at −50°C/min
	7. Storage (insert into bank)	0.25 M then 0.125 M sucrose	Plunge straws into LN_2 (~-196°C) for storage
	8. Thaw (remove from bank)	TS/DS/WS above	Remove straw; thaw rapidly by exposure to ambient temperature
	9. Stepwise dilution / rehydration	T1 → T2 → T3	5 min each at room temperature: T1 → T2 → T3 → return to culture medium
	10. Return to culture		Transfer into equilibrated culture

			medium at 37°C for assessment and transfer

4.3 Results for embryos (survival, pregnancy outcomes)

Post-thaw survival outcomes were reported using both embryo-level survival and blastomere-level survival rates. Using the modified one-step method produced a significantly higher embryo survival rate, going from 83.1% with the conventional two-step method to 86.9%. It also led to a much higher blastomere survival rate, jumping from 76.5% to 81.0% with the new one-step method [5]. After the overnight culture process, ~69% of embryos resumed cleavage in both groups, but there was not a large difference between the two methods. Most of the thaw cycles resulted in embryo transfer, and differences in the outcomes between the two methods were not due to any standout reason. Clinical pregnancy rate per transfer was 29.5% for the one-step method versus a 34.0% rate with the two-step method, and live birth rate per transfer for the modified method was 20.7% versus the traditional method being 21.8% [5]. While it was noted that there were fewer cancelled thaw cycles in the one-step method it did not lead to a difference when it comes to embryo transfer, so it ultimately did not have much of an effect in that department. Overall, the benefits to using the modified one-step method can be seen in the increased embryo survival rates, and while the two-step method can be useful, this new method can increase efficiency with both time and manpower, allowing for similar if not better survival rates and does not compromise on normal clinical outcomes.

Results:

Overall, embryo cryopreservation outcomes are commonly judged by its post-warming survival, its use for clinical pregnancy or live birth, and health of the newborn when the birth is complete.

The process of cryopreservation has mainly shifted toward vitrification because it is better designed to avoid ice crystal formation than slow freezing through rapid cooling and higher CPA loading, and it goes hand in hand with controlled osmotic recovery during warming. In the slow-freezing method, embryos are cryopreserved using PROH and sucrose solutions with programmable cooling and manual seeding, which results in high post-thaw survival and measurable clinical success, and a modified one-step CPA process can improve survival metrics relative to the conventional two-step exposure. However, that same process does not drastically improve clinical pregnancy and live birth rates, and in fact remains quite similar between slow-freezing variants. In contrast, the vitrification protocols show consistently increased post-warming survival across embryo stages and support strong clinical performance using those methods. They also illustrate that outcomes depend on both the vitrification step (CPA composition and carrier handling) as well as the warming step (sucrose-based dilution design). Taken together, these findings reinforce the current landscape of cryopreservation, that modern cryopreservation success is driven not only by choosing between the two protocols of vitrification versus slow freezing, but also by optimizing the solution compositions and step sequences that work to control dehydration, prevent ice injury, and enable safe rehydration. Ultimately, all of these changes and processes are all to work towards reliable embryo availability for transfer and the broader goals of embryo banking within IVF.

5. Final Conclusion:

Cryopreservation methods of sperms, oocytes, and embryos vary greatly due to the basic biologic variation between these cell types. These variations have a direct impact on the selection of the cryoprotective factors or agents, freezing procedures, and ultimate clinical outcomes reported in assisted reproductive technology (ART). When contrasting their biological properties, methodological methods, and findings on survival, one can easily figure out why each of these strategies needs various approaches.

Cell size is among the most significant differences, as far as biology is concerned. Spermatozoa are highly streamlined and extremely small with low cytoplasm, and the ratio of area to volume of their bodies is very high. This structural feature permits water and cryoprotectants to pass through the membrane comparatively faster, thus eliminating the chances of the formation of intracellular ice during the freezing process. Human oocytes, on the contrary, are some of the largest cells in the body with a diameter of about 120 micrometers. The large volume of their cytoplasmic space and extremely low ratio of surface to volume make rapid dehydration challenging, thus making them more susceptible to intracellular ice formation. Embryos are located between these two extremes. Although individual blastomeres may be smaller than oocytes, the embryos during cleavage or blastocytic phase do have a large amount of intracellular water and an even fluid-filled cavity (blastocoel), thus making it even more sensitive to the formation of ice crystals.

The three subjects also vary in the permeability of the membrane. The permeability of the membranes of sperms to glycerol and other penetrating cryoprotective agents (CPAs) is relatively low, and it is possible to dehydrate sperms and protect them in the intracellular environment.

Oocytes, in their turn, exhibit comparatively low membrane permeability, which implies that water gets out very slowly and CPAs get in very slowly. This allows only a small margin within which dehydration has to take place without bringing about osmotic shock or cytotoxicity. Embryos are more tolerant to stages, although, on the whole, they show less tolerance than oocytes and more tolerance than sperm. Osmotic stress and cryoprotectant toxicity due to the presence of large amounts of water and low permeability cause oocytes to be highly sensitive.

These cells are further characterized as structurally sensitive. During freezing and thawing, sperm are mostly susceptible to membrane damage, oxidative stress, mitochondrial dysfunction, and DNA fragmentation. The oocytes have a very sensitive meiotic spindle of chromosome arrangement that is also very temperature sensitive. Temporary exposure to sub-physiological temperatures can affect the integrity of the spindle, resulting in misaligned and possibly aneuploid chromosomes. At least in cases of early development stages, it is sensitive to damage not only of the physical aspects of embryos, but also other insidious aspects of environmental changes that could change epigenetic regulation patterns. These biological differences are the reason cryopreservation procedures need to be in place with regard to each type of cell.

Glycerol-based slow freezing is the conventional method of cryopreservation of sperm methodologically. The initial successful application of a highly effective permeating cryoprotectant was glycerol, which was successfully used in human sperm freezing and is still extensively used due to the relative sperm tolerance to osmotic changes and exposure to CPA. The glycerol concentrations needed are usually 5-10% to prevent the intracellular ice formation but retain acceptable post-thaw motility. Despite the studies being conducted and the potential of sperm vitrification that allows quick processing and less ice crystal structure, the advantages of the technology in comparison with traditional freezing are not necessarily uniform, and manipulation is essential. Consequently, the use of glycerol-based slow freezing has been proven to be consistent and established to be used in regular sperm banking.

On the contrary, vitrification is now favored with regard to oocytes and embryos. Slow freezing goes via some ranges of intermediate temperatures at which ice crystals may develop, causing severe hazard to large, water-filled cells. The Oocytes are particularly susceptible to this since the presence of ice inside cells or spindle destabilization may permanently affect the integrity of chromosomes. In vitrification, the presence of ice is completely avoided through excessive concentrations of penetrating cryoprotectants (ethylene glycol and dimethyl sulfoxide, typically at concentrations of about 15 per cent or higher) and extremely high cooling and warming rates. The swift change into a glass-like condition will not allow crystallization. In the case of embryos, vitrification also enhances survival of the embryo after warm-up, as opposed to slow freezing, especially during the blastocystic stage.

In comparison of the results, vitrified embryos usually had the highest post-warming survival tendencies, usually more than 90. Vitrified oocytes also exhibit survival rates of close to 90-95, which, compared to the previous methods of slow-freezing the oocyte, was far better, with survival rates of around 60. Existence of sperms following freezing is commonly determined according to motility and membrane viability, and not structural endurance. Although motility reduces following thawing, fertilization rates, and especially when applying ICSI, are usually similar to those obtained through fresh sperm. Therefore, sperm cryopreservation can be found to be clinically effective even though the motility is reduced.

Vitrified blastocysts have the highest implantation and cumulative pregnancy rates in the contemporary practice of IVF. Transfer cycles of frozen embryo transfer often have the same or even slightly higher outcomes than fresh transfer cases. Vitrified oocytes may also lead to successful fertilization and pregnancy, particularly in case eggs have been frozen at a younger age. Cryopreservation of sperm enhances high rates of pregnancy using the IVF and ICSI

methods, especially since the latter involves the use of a single viable sperm per oocyte. In spite of the fact that the majority of children born out of cryopreservation of gametes or embryos are healthy, the current research keeps following some cases of rare risks, including imprinting diseases.

On the whole, sperm are the most biologically forgiving to freeze because of their minuscule size and membrane permeability, and, thus, glycerol-based slow freezing can be used. Oocytes are the most delicate due to their large size, the high well content of water, and the sensitivity of the spindle; hence vitrification conditions are vital in the successful preservation of oocytes. Presently, embryos, especially at the blastocyst stage have the best survival and pregnancy rates after being vitrified. The development of slow freezing and vitrification of oocytes and embryos constitute one of the most prominent contributions to the assisted reproductive technology, which shows how biological peculiarities predetermine the selection of methods and eventually defines clinical results.

5.1 Future Directions

Cryopreservation in assisted reproductive technology (ART) is still developing and investigators and clinicians aim to enhance survival, minimize cellular injuries and speed up and simplify the tasks of the laboratory. Despite the fact that modernized techniques like vitrification and slow freezing have enabled the preservation of sperm, oocyte, and embryo to be an extremely successful activity, technological progress can be achieved to a substantial degree. The trend is likely to be enhanced with future discoveries through advancements in the formulations of the cryoprotectants, which can be more automated, and also, the overall safety and reliability of the cryostorage systems.

The development of new cryoprotectants and combinations of cryoprotectants is one of them. The classical methods of cryoprotectants as glycerol, dimethyl sulfoxide (DMSO) and ethylene glycol successfully prevent ice assemble in the cell, however, when high stakes are employed, such instruments can also imply cytotoxicity and osmocrisis. Scientists are already looking into the use of other cryoprotectants as natural sugars, polymers, antifreeze proteins and plant-derived compounds that can be equally protective and less toxic. As an example, molecules that are based on antifreeze proteins in cold-adapted organisms can potentially inhibit ice formation and reduce chemical damage in the reproductive cells. Antioxidants and protective additives like melatonin, glutathione and vitamin E are also under research on their potential to decrease oxidative damage in freezing and thawing.

Other opportunities include automating and standardizing the cryopreservation processes. Nowadays, numerous vitrification processes are very manual-intensive and they depend on embryologists and this may create variation across lab to lab and even operator to operator. The latest automatized systems of vitrification are designed to better control the exposure duration, the concentration of different additives in the cryoprotectant mixture, and the rate of cooling as possible methods to eliminate human error and enhance the reproducibility. Programmable cryogenic devices and robotic liquid handling systems are also to be invented to standardize the freezing and warming of steps. The technologies might ensure that the cryopreservation process remains more uniform among clinics, as well as enable laboratories to process a greater number of samples with elevated efficiency and control the quality.

The other promising technology which could better the cryopreservation process is the advancement in microfluidic technologies. Microfluidic devices have the ability to handle very low fluid volumes with a high degree of accuracy and in this way, cells can be gradually exposed to cryoprotectants with controlled exposure and reduced osmotic shock. These systems could be of special importance when dealing with sensitive cells like the oocytes and embryos where a slight variation in the composition or the temperature of the solution can result in cell injury. Also,

microfluidic systems might allow keeping very small sperm counts alive, which is significant to patients with severe male infertility or surgical obtained sperm.

It is also the focus of future trends in reproductive cryobiology to enhance the laboratory safety and control of contamination. Conventional systems of open vitrification process permits direct exposure between samples and liquid nitrogen which may also permit microbial contamination. Consequently, most laboratories are moving to forms of closed vitrification that physically separate samples yet having high cooling rates. There is also the use of alternative cryostorage containers, sterile liquid-based nitrogen filtration and the introduction of monitoring to maintain sample safety over a period of time.

Lastly, the increased development of data monitoring and artificial intelligence can also be involved in more streamlined cryopreservation protocols. Big data provided by fertility clinics can be problem-solved with the help of machine learning and identify the variables affecting cryosurvival, embryo viability, and pregnancy rates. This incorporation of lab metrics, environmental factors, and patient features can assist AI-based systems to optimize cryopreservation regimes in addition to making the best treatment plans to suit discrete patients.

On the whole, the advances in the field of cryopreservation in the future are likely to be integrated by the enhancement of the best methods of cryoprotectants chemistry, automation of the laboratory, microfluidic drug delivery, and computerized monitoring programs. Such innovations will be directed to a reduction of cellular damage, enhancement of reproducibility, as well as an augmentation of the safety and success rates of fertility preservation methods in the area of assisted reproductive medicine.

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